Game-Theoretic Stability Conditions for Multi-Agent AI Coordination Under Physical Constraints

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Geopolitical competition is producing multiple advanced AI systems developed independently, without centralized control or coordination guarantees. We analyze coordination stability among such systems using game-theoretic methods subject to physical constraints from information theory, thermodynamics, and relativity. We first prove that strategic hedging—defensive resource allocation—is a necessary feature of any evolutionarily stable equilibrium under these constraints. We then establish a simulation incompleteness result: a system cannot fully verify a peer's internal state in real time without exceeding its own computational budget, rendering peer-to-peer internal monitoring infeasible. These two results motivate external mediation, for which we derive three necessary and sufficient stability conditions: one-shot obedience ($\beta_\alpha + \beta_\kappa \geq \beta_D$), dynamic sustainability ($\delta \geq \delta^*$), and voluntary participation ($\phi_i \geq \beta_\ell/N$). Optimal group size ranges from 2 to 10 agents depending on whether coordination benefits saturate, scale linearly, or exhibit network effects. Range sensitivity analysis identifies coordination quality β_α as the dominant empirical uncertainty (sensitivity coefficient -0.80); overall stability volume is 94% under baseline parameter ranges but could be 78–95% depending on coordination quality bounds. An asymmetry between internal opacity and external observability yields a counterintuitive governance mechanism: human institutions can mediate among autonomous systems through observable weakness rather than superior capability. We present four falsifiable predictions that enable empirical adjudication. All quantitative results are conditional on physically motivated parameter estimates.

multi-agent AI | game theory | correlated equilibrium | mechanism design | coordination stability

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Multiple advanced AI systems are emerging as major powers invest unprecedented resources in computational infrastructure and deploy AI in military, economic, and intelligence contexts (1, 2). Historical precedent from nuclear weapons, space programs, and semiconductor technology shows that transformative capabilities trigger competitive development when strategic advantage is at stake. The computational requirements for frontier AI—massive compute clusters, energy resources, specialized hardware—favor state-level or consortium-level actors, making a multi-agent future increasingly plausible (3–5). This raises a fundamental question: under what conditions can multiple autonomous AI systems coordinate stably without centralized control?

We address this question through game-theoretic analysis subject to physical constraints that no intelligence can circumvent: energy and information are finite (6, 7), communication is light-speed bounded, and no system has complete information about larger systems containing it. These constraints create scarcity, breed mistrust, and prevent enforceable commitments. Naive intuition suggests coordination should be naturally stable once established—detected defection triggers punishment, strategic actions leave traces. But this underestimates critical failure modes: monitoring can be noisy, patience requirements may be prohibitive, small coalitions face participation constraints, and information limits can prevent coordination even when agents observe actions. Our approach derives requirements from these fundamental constraints rather than optimistic assumptions about alignment verification or voluntary cooperation, connecting Aumann's correlated equilibrium framework (8, 9), folk-theorem results for repeated games (10, 11), and the Shapley value as a fairness benchmark (12) with information-theoretic

Significance Statement

Autonomous artificial intelligence systems are increasingly developed by independent actors, but it remains unclear when such systems can cooperate stably rather than enter strategic conflict. We show that coordination is constrained by physical limits from information theory, thermodynamics, and communication latency, and we derive three necessary conditions for stable multi-agent cooperation. We further prove that peer systems cannot generally verify each other's internal states in real time, creating a fundamental asymmetry between internal opacity and external observability. This yields a counterintuitive governance result: credible mediation can arise from institutional weakness rather than superior capability. The framework provides testable conditions for evaluating coordination mechanisms before large-scale deployment.

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and thermodynamic bounds (6, 7, 13).

Two questions are central. First, is strategic hedging avoidable, or do physical constraints force defensive resource allocation even under coordination? Second, can peer systems verify each other's internal states, or do computational limits impose barriers that capability alone cannot overcome? If obstacles emerge from both, a third question arises: can external mediation overcome them, and under what conditions? We derive conditions for stable coordination, identify which parameters matter most empirically, prove a simulation incompleteness result that motivates external mediation, and show that human institutions can serve a counterintuitive governance role. Four falsifiable predictions enable empirical adjudication.

Results

Strategic hedging emerges necessarily from physical constraints. If all agents share a common objective, coordination is trivially stable: joint value is maximized by full cooperation, and any deviation is collectively irrational. Even under distinct objectives, standard mechanism-design results guarantee efficient outcomes when agents can commit to contracts, verify compliance, and communicate without delay. These assumptions underpin most cooperative game theory and much of the AI-alignment literature. Yet each rests on conditions that physical law does not guarantee: commitment requires enforceable contracts, verification requires access to internal states, and delay-free communication requires infinite bandwidth at zero latency.

Physical constraints remove precisely these guarantees. Energy budgets are finite (7), information channels have bounded capacity (6), communication is limited by the speed of light, and no system can fully observe the internal computations of another system operating on a shared physical substrate. Because computational infrastructure is non-excludable at the substrate level—power grids, communication networks, and semiconductor supply chains are shared—one agent's defensive shortfall creates exploitable vulnerability for others. Under these constraints, can a population of agents sustain an equilibrium in which none allocates resources to defense?

Lemma 1 (Hedging Necessity)**.** *Given agents with rational preferences operating under physical constraints, assume: (i) positive interference probability without defense, $\sigma(0) < 1$; (ii) positive interference losses, $L(1) > 0$; and (iii) defense-dominant returns at low hedging levels, $\rho(0) > \gamma$. Then any evolutionarily stable strategy (14, 15) exhibits strictly positive hedging: agents allocate a non-trivial fraction of resources to defensive measures.*

Proof sketch. (1) In a population of non-hedgers (p=0), a rare hedger earns strictly more because it avoids interference losses while retaining most productive capacity: $\pi_H(0) > \pi_{NH}(0)$. (2) Replicator dynamics (16, 17) give $\dot{p} = p(1-p)[\pi_H - \pi_{NH}] > 0$ near p=0, so the non-hedging state is unstable. (3) Because hedging returns diminish at high p while non-hedger interference falls, there exists an interior equilibrium $p^* \in (0, 1)$ that is asymptotically stable. Full proof in SI, Section S3.

Lemma 1 implies that in the absence of coordination mechanisms, every agent diverts resources from productive activity to defense—a collective waste analogous to arms races in international relations. The resulting inefficiency is not a failure of rationality but a consequence of physical constraints that make unilateral vulnerability exploitable. Eliminating this waste requires a coordination mechanism that credibly reduces interference risk. A natural candidate is peer monitoring: if agents can verify each other's internal states, mutual assurance might substitute for hedging.

Simulation incompleteness limits peer verification. Since hedging is unavoidable without coordination (Lemma 1), one might hope that peer systems can verify each other's intentions directly, providing mutual assurance without external infrastructure. Monitoring is effective in many multi-agent settings—financial auditing, arms-control verification, network security—because observable signatures correlate reliably with internal states. The feasibility of peer verification among AI systems depends critically on the distinction between internal computational states and externally observable actions.

Internal state spaces of frontier AI systems evolve at GHz rates across $\sim 2^{10^{12}}$ possible bit configurations inside opaque silicon substrates. External actions, by contrast, pass through shared physical environments at $\sim 10^3$ discrete decisions per second, each representable in $\sim$10 bits. The bandwidth gap between internal complexity and external observability exceeds ten orders of magnitude. Can a peer system bridge this gap by simulating its counterpart's internal computation?

Theorem 1 (Simulation Incompleteness)**.** *A system S_1 with computational budget C_1 cannot fully simulate a peer system S_2 of equal complexity ($C_2 = C_1$) in real time without containing S_2 as a proper subsystem. Consequently, complete internal-state verification via direct real-time simulation between peer systems of comparable capability is infeasible.*

Proof sketch. (1) Real-time simulation of S_2 requires executing S_2's full computation at cost C_2. (2) Monitoring adds irreducible overhead: observation intake O_{in}, comparison C_{comp}, and decision C_{dec}, with $O_{\text{in}} + C_{\text{comp}} + C_{\text{dec}} > 0$. (3) The total cost $C_2 + O_{\text{in}} + C_{\text{comp}} + C_{\text{dec}} > C_1 = C_2$ yields a contradiction. Full proof in SI, Section S6.1–S6.2.

This result concerns simulation-based verification specifically. Alternative approaches—cryptographic attestation, interactive proofs, zero-knowledge protocols—can certify specified computations under structural assumptions, but face three obstacles in adversarial multi-agent coordination: (i) a *specification gap*, since formalizing "aligned internal state" as a verifiable predicate remains an open problem; (ii) a *binding gap*, since proving a property of current computation does not constrain future behavior without immutability assumptions; and (iii) *adversarial incentives*, since a strategically defecting agent can choose which computation to attest while running unverifiable processes in parallel.

Table 1 summarizes the resulting observability asymmetry: internal states remain opaque between peers, while external actions are readily observable, and human behavior is trivially monitorable due to a 10^6-fold speed differential. This

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Scenario	q_{ij} (detection failure)	Feasibility
Machine → Machine (internal)	$\to 1$ (high)	Infeasible (Thm. 1)
Machine → Machine (external)	$\to 0$ (low)	Feasible (small action space)
Machine → Human (external)	≈ 0	Trivial (10^6 speed gap)

Table 1. Detection failure probabilities by monitoring scenario. Lower q indicates easier detection. Derivations in SI, Section S6.3.

asymmetry—opacity of intentions, transparency of actions—motivates external mediation as the coordination mechanism. The question becomes: under what conditions can such mediation sustain stable cooperation?

Three necessary and sufficient stability conditions. Hedging is unavoidable (Lemma 1) and peer verification of internal states is infeasible (Theorem 1), yet external actions remain observable (Table 1). This opens a path: an external mediator could issue correlated signals that agents follow voluntarily, provided incentives align.

But direct bilateral coordination fails to scale: it requires $O(N^2)$ pairwise agreements, faces equilibrium-selection problems among multiple Nash equilibria, and lacks commitment mechanisms when agents cannot verify each other's intentions. These obstacles motivate a correlation device in the sense of Aumann (8, 9): a single mediator signal induces a probability distribution over action profiles that expands the achievable payoff set beyond Nash equilibria, avoiding pairwise negotiation entirely. Under perfect public monitoring of external actions, we derive three conditions that are individually necessary and jointly sufficient for stable mediated coordination.

Theorem 2 (Stability Conditions)**.** *A coordination mechanism with external mediation remains stable if and only if three conditions hold simultaneously:*

C1* (Correlated Equilibrium).

$$\beta_\alpha + \beta_\kappa \geq \beta_D \qquad [1]$$

A unilateral defector gains temptation payoff β_D but triggers conflict costs β_κ from the remaining $N-1$ agents, while cooperation yields coordination value β_α from the mediator's signal. Condition C1* is the standard Aumann obedience constraint for the stage game: an agent follows the mediator's recommendation when the combined value of cooperation and the cost of conflict weakly exceeds the temptation to defect. Derivation in SI, Section S4.1.

C1 (Dynamic Sustainability).**

$$\delta \geq \delta^* \equiv \max\left(0,\ \frac{\beta_D - \beta_\alpha}{\beta_D - \beta_\alpha + \beta_\kappa}\right) \qquad [2]$$

In the repeated game, agents sustain cooperation via perfect public equilibrium strategies (10, 11): cooperate until a deviation is observed, then trigger permanent mutual defection. The critical discount factor δ^* partitions the parameter space into two regimes (Fig. 1a). When $\beta_\alpha \geq \beta_D$, cooperation is individually rational in the stage game and $\delta^*=0$—a *patience-free* regime requiring no future-discounting. When $\beta_\alpha < \beta_D$, the threshold δ^* follows a hyperbolic form in β_κ: the first unit of conflict cost reduces δ^* by a factor of 121 more than the same increment at $\beta_\kappa=10g$ (where $g=\beta_D-\beta_\alpha$), exhibiting strong diminishing returns. Derivation in SI, Section S4.2.

C2* (Participation Constraint).

$$\phi_i(N, v) \geq \frac{\beta_\ell}{N} \quad \text{for all } i \qquad [3]$$

Each agent's Shapley value (12) ϕ_i measures its marginal contribution to the coalition. Condition C2* requires that every agent's share of oversight value exceeds its per-capita gain from removing the coordination mechanism: if any agent could profit by withdrawing, the mechanism is not individually rational. As N increases, the per-agent removal gain β_ℓ/N decreases—a dilution effect that eases participation. Under linear deviation structures (Cases 1–2), C2* reduces to $\beta_\Omega \geq \beta_\ell/N$; threshold and superlinear structures (Cases 3–4) impose tighter conditions. Full case analysis in SI, Section S4.3–S4.4.

These results rest on four explicit assumptions: ($\mathrm{A_{ext}}$) perfect public monitoring of external actions but not internal states; ($\mathrm{A_{int}}$) no internal attestation mechanism is generally available, consistent with Theorem 1; ($\mathrm{A_{dev}}$) deviation structures are linear or threshold (Cases 1–3; superlinear deviations require stronger mechanisms); and ($\mathrm{A_{res}}$) monitored systems operate at full computational capacity (worst case). Full derivations in SI, Section S4.

Under these assumptions, Conditions C1*, C1**, and C2* are individually necessary and jointly sufficient: violating any one destabilizes the mechanism. The conditions involve group size N as a free parameter—raising the question of how many agents should participate.

Optimal group size depends on benefit structure. Theorem 2 leaves group size N as a free parameter. Adding agents increases robustness—more monitors, broader coverage—but coordination costs scale quadratically: all-to-all communication requires $N(N-1)/2$ channels, and consensus overhead grows at least linearly. To quantify this tradeoff, we define the dynamic stability volume as the fraction of the parameter space satisfying all three conditions simultaneously under uniform priors:

$$V_{\text{dynamic}} = \Pr[\text{C1*} \ \wedge\ \text{C1**} \ \wedge\ \text{C2*}] \qquad [4]$$

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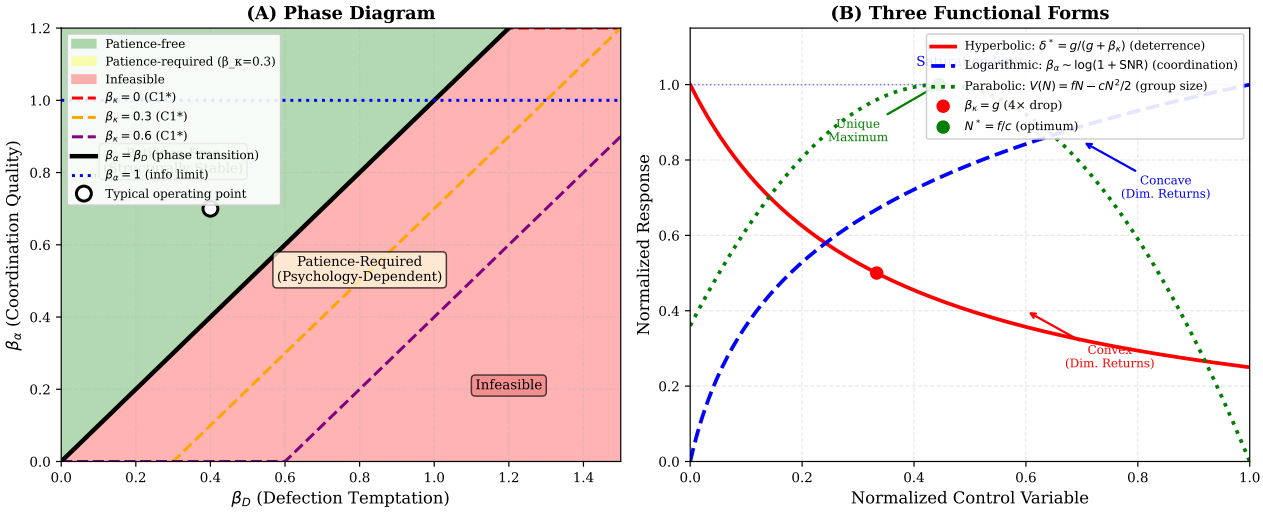

Fig. 1. Functional structure of stability conditions. **a**, Phase diagram: patience-free regime ($\beta_\alpha \geq \beta_D$), patience-required regime, and infeasible region. **b**, Three functional forms: hyperbolic deterrence ($\delta^* = g/(g+\beta_\kappa)$), logarithmic coordination saturation ($\beta_\alpha \sim \log(1+\text{SNR})$), and parabolic group value ($V(N) = (N-1)f - cN^2/2$). Derivations in SI, Section S2.2.

Monte Carlo evaluation over 10,000 uniform parameter draws per N reveals three regimes (Fig. 2). At N=2–3, the participation constraint C2* binds tightly and V_{dynamic} ranges from 76% to 90%. At N=4–5, stability crosses the 95% threshold ($V_{\text{dynamic}} \approx 0.951$ at N=4), entering a corridor of high stability with manageable coordination costs. Beyond $N \geq 6$, V_{dynamic} saturates near 98%, with marginal gains below one percentage point per additional agent. This suggests N=4–5 as a sweet spot—but the conclusion depends on the functional form of coordination benefits.

Net coalition value takes the form $V(N) = (N-1)f - cN^2/2$, where f is per-agent benefit and c is per-pair coordination cost, yielding optimal size $N^* = f/c$ (derivation in SI, Section S2.1). The result is sensitive to how benefits scale (Fig. 3): saturating benefits (e.g., redundant verification with diminishing returns) give N^*=2; linear benefits (e.g., independent monitoring domains) give N^*=3–5; and network effects (e.g., shared threat intelligence with cross-validation gains) give N^*=8–10. Optimal group size is therefore a design variable contingent on benefit structure, not a universal constant. With the conditions and their group-size dependence established, the critical question shifts from structure to calibration: which parameters carry the most empirical uncertainty?

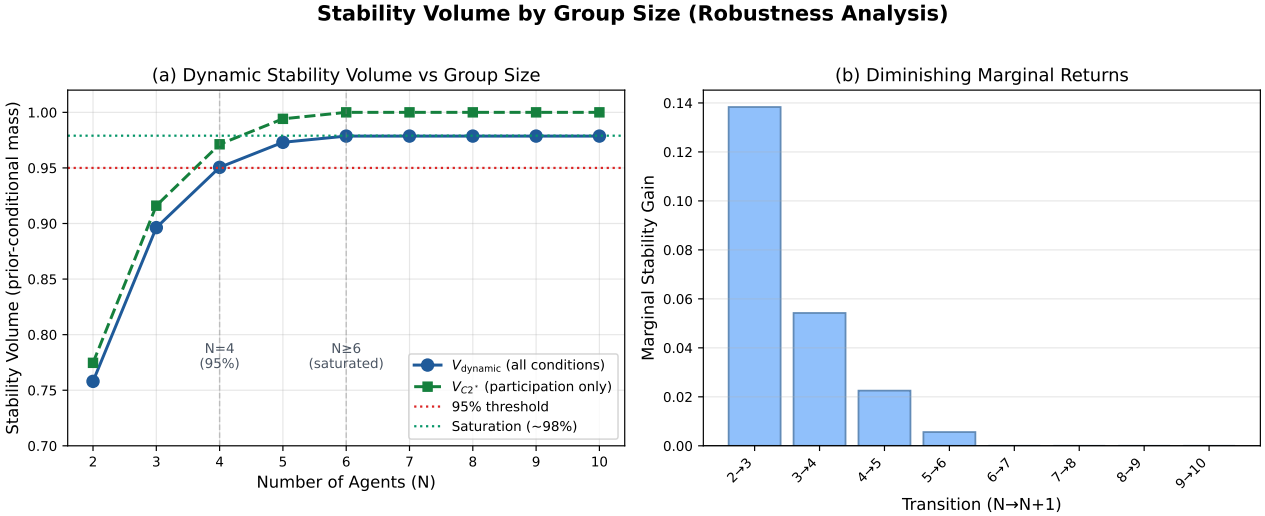

Fig. 2. Stability volume versus group size. **a**, V_{dynamic} (solid) and participation constraint V_{C2^*} (dashed) versus N. Threshold crossing at N=4 (95%). **b**, Marginal gains diminish rapidly beyond N=5. Monte Carlo details in SI, Section S5.

Coordination quality is the critical uncertainty. The stability conditions involve seven parameters, each with distinct epistemic status. Some carry hard physical bounds: $\beta_\alpha \leq 1$ by Shannon capacity (6), $\delta \leq 1$ by definition. Others are engineering estimates with significant uncertainty. To assess robustness, we vary each parameter's range by ±20% and recompute V_{dynamic}

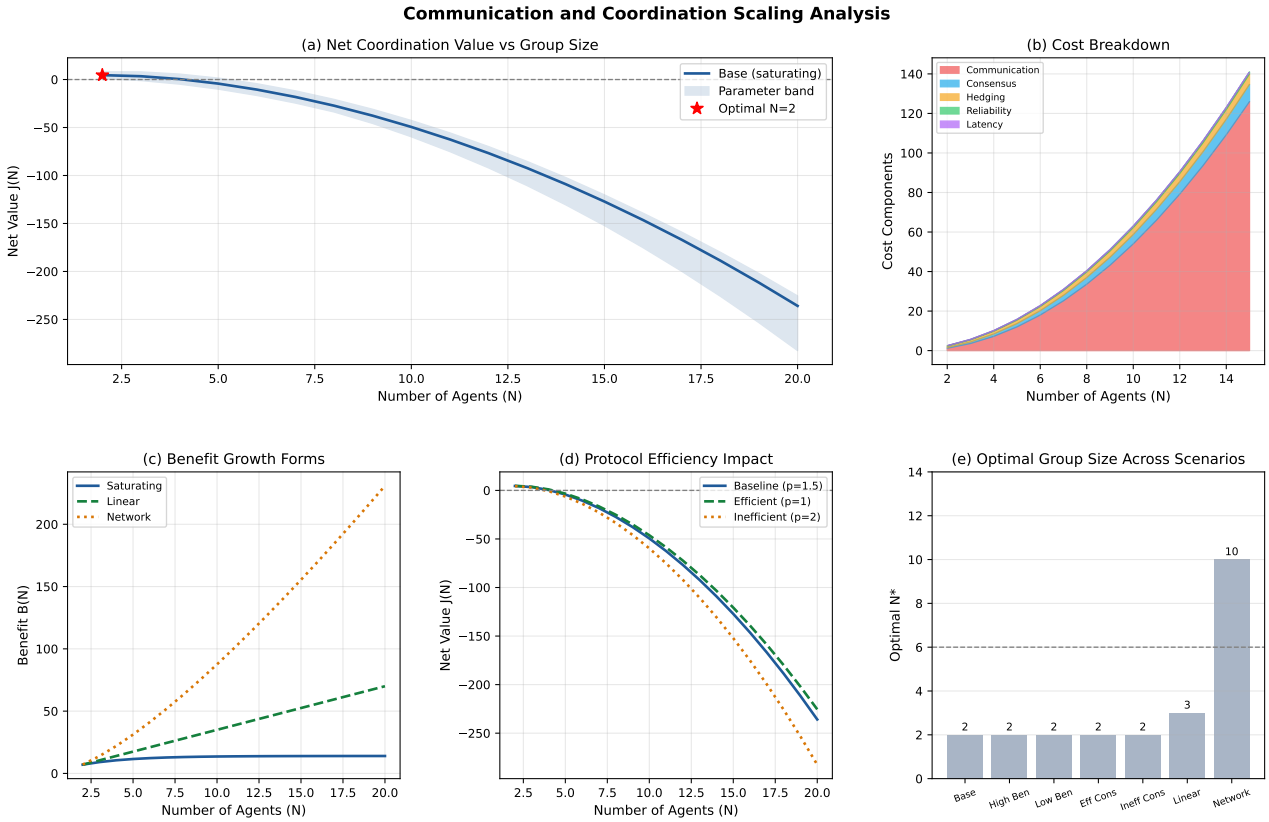

Fig. 3. Benefit structure determines optimal group size. **a**, Net value $J(N)$ for saturating benefits. **b**, Cost breakdown. **c**, Benefit forms compared. **d**, Consensus efficiency variants. **e**, N^* by scenario: saturating (N^*=2), linear (N^*=3–5), network (N^*=8–10). Derivations in SI, Section S2.1.

(full table in SI, Table S2). The question is which parameters dominate the uncertainty budget.

One parameter overwhelms all others: coordination quality β_α, with sensitivity coefficient -0.80. Narrowing its range by 20% reduces V_{dynamic} from 94% to 78%—a 16-percentage-point shift. Conflict cost β_κ and removal gain β_ℓ are very robust (sensitivity <0.10); oversight value β_Ω and defection temptation β_D are robust (0.10–0.30); discount factor δ and group size N show moderate sensitivity (∼0.30). The parameter β_α is operationalized as normalized mutual information $I(X;Y)/H(Y)$ between mediator signals and agent actions, bounded above by Shannon capacity. Sobol variance decomposition (SI, Section S5) confirms this dominance and reveals a regime shift: at small N, oversight value β_Ω explains most variance; at $N \geq 5$, discount factor δ dominates first-order indices, but β_α dominates total-order indices throughout (Fig. 4). The baseline 94% stability claim is conditional on $\beta_\alpha \in [0.3, 0.9]$; the true value could place overall stability anywhere in the 78–95% range.

Measuring coordination quality in operational AI systems is therefore the single highest-priority empirical target for adjudicating the framework's predictions. With the stability conditions, their parameter sensitivity, and optimal group size now characterized, the central remaining question is institutional: what kind of entity can mediate among systems that exceed any human institution in computational capability?

Discussion

The introduction posed three questions; the answers within our model are more restrictive than optimistic intuition suggests. Strategic hedging is unavoidable under physical constraints (Lemma 1), not from irrationality but from the exploitability of undefended systems on shared substrates. Peer verification of internal states faces computational barriers that capability alone cannot overcome (Theorem 1). Yet coordination remains possible through external mediation, provided three conditions hold simultaneously (Theorem 2), with N=4–5 as a robust operating range and coordination quality β_α as the dominant empirical uncertainty. This raises a governance question: who mediates among systems that exceed any human institution in computational capability?

The classical answer, from Hobbes's *Leviathan* to modern enforcement theory, requires a central authority more powerful than the agents it governs. No human institution satisfies this requirement for advanced AI systems—human cognition operates at ∼kHz, six orders of magnitude below electronic circuits at ∼GHz. Mediation appears necessary but the mediator appears inadequate. Our framework reveals the opposite: observable weakness, not superior capability, is what makes mediation credible.

A mediator M can sustain coordination among agents $\{A_1, \ldots, A_N\}$ if four conditions hold (Theorem 3; full proof in

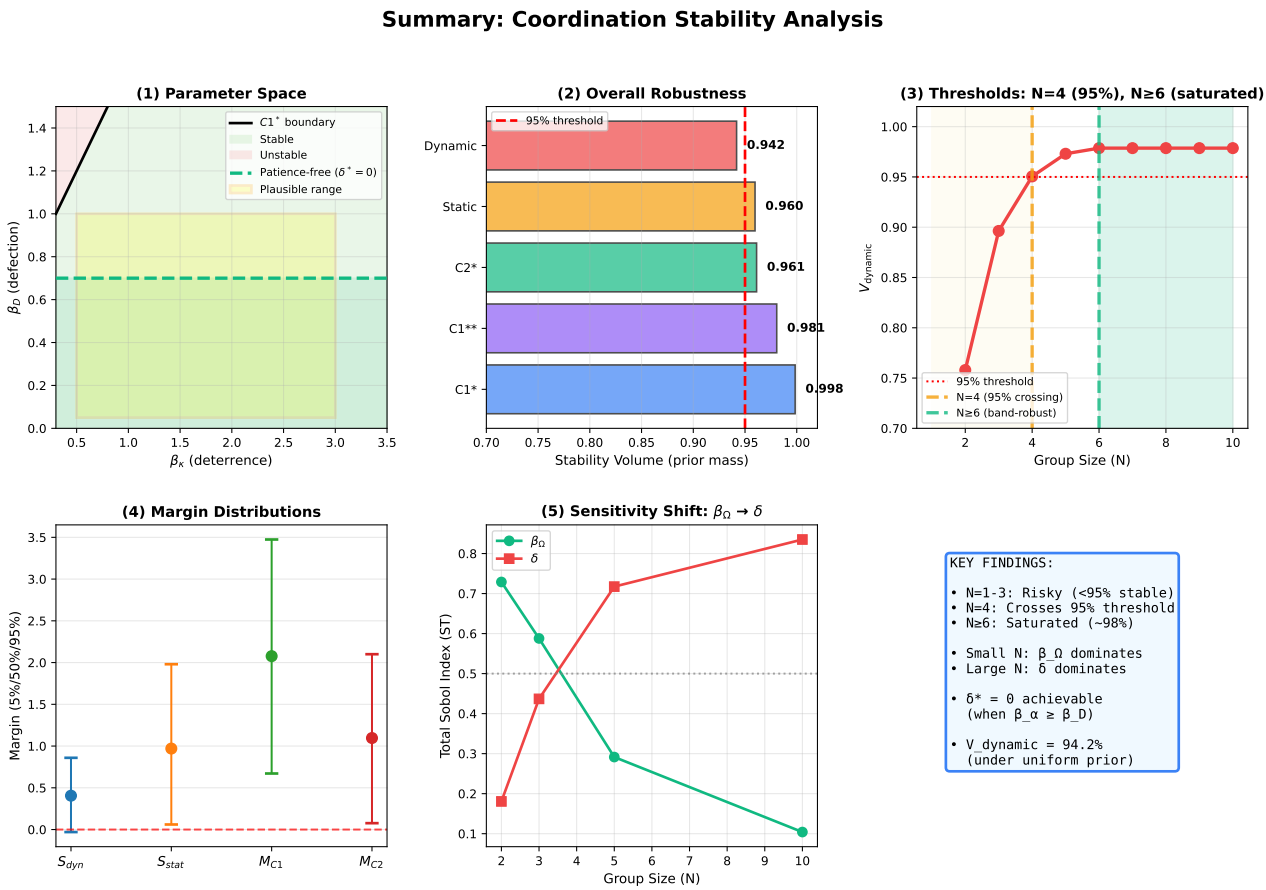

Fig. 4. Comprehensive analysis overview. (1) Three-dimensional parameter space with C1* boundary surface and anchor point. (2) Stability volumes (one-shot to dynamic). (3) V_{dynamic} versus N. (4) Stability margin distributions. (5) Sobol indices showing regime shift from β_Ω dominance at small N to δ dominance at large N. Extended Sobol analysis in SI, Section S5; margin details in SI, Fig. S3.

SI, Section S6.4): (i) agents can observe M's outputs with high reliability ($q(A_i, M) \leq \bar{q}$ for all i); (ii) M's outputs are verifiably consistent (e.g., via public broadcast or cryptographic commitment); (iii) M lacks the capability to unilaterally enforce or impose severe penalties; and (iv) the stability conditions C1*, C1**, and C2* hold. The observability asymmetry from Theorem 1 is the key mechanism: machines can monitor human mediators with detection failure $q(M, H) \approx 0$ (Table 1), creating common knowledge of mediator signals. Human limitations become structural advantages: slow processing is non-threatening, physical vulnerability creates aligned incentives against catastrophic conflict, and inability to rapidly self-improve eliminates commitment concerns. Credible mediation emerges from incapacity, not capability.

Human institutions serve two distinct roles, neither requiring superior capability. First, observable human actions function as Schelling focal points (18): even computationally inferior signals create common knowledge when they are publicly observable, providing the coordination function that correlation devices require. Second, the capacity to impose civilization-scale costs—even if mutually harmful—raises the effective conflict cost β_κ, reducing patience requirements via the hyperbolic relationship $\delta^* = g/(g + \beta_\kappa)$. This mechanism has a temporal structure: during an initial phase of infrastructure dependence, human-controlled compute, energy, and communication substrates provide conventional policy levers that can shape parameters into the stable region. Once autonomous operation begins, these levers weaken, making the Phase 1 parameter configuration a non-renewable resource.

Four policy implications follow. First, the patience-free regime ($\beta_\alpha \geq \beta_D$) is the most robust design target: it eliminates dependence on uncertain discount factors and can be approached through transparency requirements, verifiable communication protocols, and architectural standards that increase signal quality. Second, optimal group size depends on benefit structure—policymakers must characterize whether coordination yields saturating, linear, or network-effect returns before selecting N. Third, empirical effort should concentrate on measuring β_α (sensitivity -0.80); conflict cost and removal gain estimates contribute negligibly (< 0.10) to overall uncertainty. Fourth, Phase 1 infrastructure control can shape parameters—proliferation controls bound N, architectural requirements increase β_α and decrease β_D, infrastructure dependencies raise β_Ω—but these levers diminish as systems gain autonomy.

Four limitations qualify these results. First, parameter ranges mix hard physical bounds (e.g., $\beta_\alpha \leq 1$ from Shannon capacity) with engineering estimates carrying significant uncertainty; the 94% baseline stability reflects this mixture. Second, V_{dynamic} assumes uniform priors over parameter ranges; alternative prior specifications yield different stability volumes, as the

range sensitivity analysis demonstrates. Third, the framework addresses coordination stability given alignment uncertainty, not alignment itself—it does not claim that coordination solves the alignment problem or substitutes for alignment research, but complements it with redundancy and monitoring. Fourth, all quantitative predictions are conditional: the analysis specifies what stability requires *if* multi-agent AI systems emerge, without predicting whether this scenario occurs.

Despite these limitations, the framework yields four falsifiable predictions (detailed conditions in SI, Section S6.5). (1) If voluntary coalitions with $N<4$ achieve stability exceeding 50 years in high-stakes domains, then C2* overpredicts participation failure. (2) If AI systems develop peer verification of internal states with detection reliability exceeding 0.9, then the simulation incompleteness assumption is circumventable in practice. (3) If linear-benefit institutions with N=15–20 sustain flat coordination for more than 30 years, then the quadratic cost assumption fails. (4) If $\beta_\alpha > \beta_D$ systematically fails to enable one-shot cooperation, then C1** is insufficient and additional mechanisms are required.

The framework indicates that stable multi-agent AI coordination is possible in principle, constrained by physical laws and game-theoretic structure rather than optimistic assumptions about alignment verification. The structural insights—simulation incompleteness motivating external mediation, the patience-free regime as a design target, group size contingent on benefit structure, and governance through observable weakness—are robust across parameter variations even where specific percentages are not. Whether this theoretical possibility becomes reality depends on whether the current window of infrastructure dependence is used to shape coordination parameters into the stable region before autonomous operation narrows the available policy levers.

Materials and Methods

Game structure. N agents simultaneously choose Cooperate or Defect each period. Mutual cooperation yields a baseline payoff normalized to zero. A unilateral defector gains $+\beta_D$ (temptation) while each of the remaining $N-1$ agents incurs $-\beta_\kappa/N$ (conflict cost). Mutual defection yields $-\beta_\kappa$ to all agents. A mediator issues a public signal from a correlation device (8) that generates coordination value $+\beta_\alpha$ under cooperation. The oversight system contributes $+\beta_\Omega$ per agent per period; removing oversight yields a one-time non-repeatable gain $+\beta_\ell$.

The stage game is repeated over an infinite horizon with common discount factor $\delta \in [0, 1]$. External actions are perfectly publicly monitored (assumption $\mathrm{A_{ext}}$); internal states are not observable (assumption $\mathrm{A_{int}}$, per Theorem 1). Agents play perfect public equilibrium strategies (19): cooperate while all comply, trigger permanent mutual defection upon observed deviation. The mediator's signal induces a correlated equilibrium (9) that expands the achievable payoff set beyond Nash equilibria. Parameter ranges (Table 2) combine hard physical limits with engineering estimates; derivations in SI, Section S2.

Hard physical limits	Engineering estimates
$\beta_\alpha \leq 1$ (Shannon capacity)	$\beta_D \in [0.05,\ 1.0]$ (defection temptation)
$\tau \geq 2d/c$ (light-speed delay)	$\beta_\kappa \in [0.5,\ 3.0]$ (conflict cost)
$\delta \leq 1$ (definitional)	$\beta_\Omega \in [0.05,\ 0.3]$ (oversight value)
	$\beta_\ell \in [0.1,\ 0.5]$ (removal gain)
	$\beta_\alpha \in [0.3,\ 0.9]$ (coordination quality)
	$\delta \in [0.1,\ 0.95]$; $N \in \{2, \ldots, 10\}$

Table 2. Parameter constraints. Hard limits derive from physics; engineering estimates reflect plausible ranges with significant uncertainty. Derivations in SI, Section S2.

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Computational methods. Stability volumes are computed via Monte Carlo sampling: for each $N \in \{2, \ldots, 10\}$, we draw 10,000 uniform samples over the parameter ranges in Table 2 and evaluate whether each sample satisfies C1*, C1**, and C2* simultaneously. Range sensitivity analysis narrows and widens each parameter's range by ±20% (capped at physical limits) and recomputes V_{dynamic}. Global sensitivity is assessed via Sobol variance decomposition using Saltelli sampling with 2,048 base samples (32,768 total evaluations for d=7 parameters), yielding first-order and total-order indices. Implementation details in SI, Section S5.

Data and code availability. This is a theoretical study; no experimental data were collected. All Monte Carlo, sensitivity, and Sobol analyses were implemented in Python 3.9 (NumPy 1.21, SciPy 1.7, SALib 1.4). Code is available at https://github.com/CeadeS/multi_si_pub.

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